6. Applications of Integrals

6.1. Volumes by Cross-Sections, Disks, and Washers

6.2.2. Solids of Revolution: The Disk method

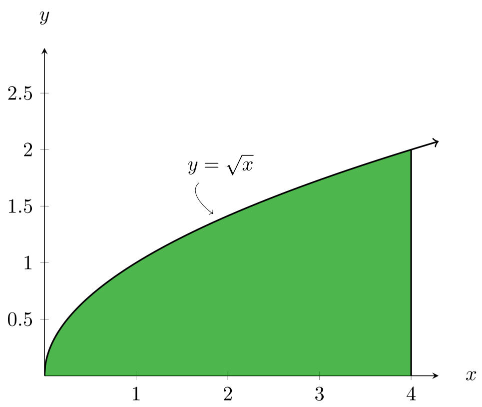
6.2b Learning Objectives:

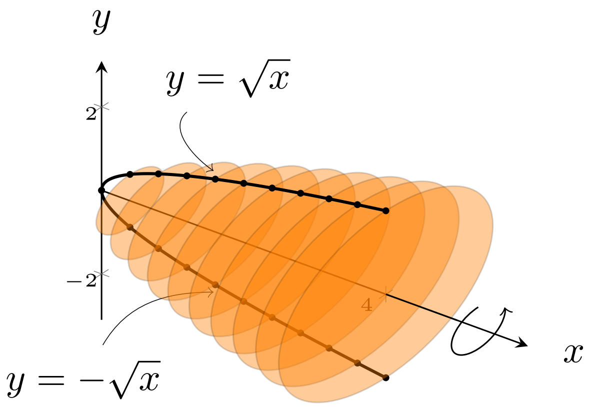
- I can use disk or washer method to find the volume of a *solid of revolution*. I can recognize that these methods are related to finding the area between curves.

Definition 6.2.1

A *solid of revolution* is the solid generated by rotating (or revolving) a plane region about an axis in its plane.

Consider the function $y = \sqrt{x}$ on $[0, 4]$. Imagine rotating it all the way around the x-axis to form a solid:





Note

If we wish to find the volume of this solid, we have only added one step to our process from before. After that we have the same type of problem as the ones with the cross-sections. We have to build the 2-D region we want to form the solid from.

1. Sketch the function and its reflection about the axis you are told to revolve around. This with the domain give you the 2-D region we go from.
2. Imagine circular cross-sections that are perpendicular to the same axis you just used.
3. Continuing as before the cross-sections will *always* be circular and have radius equal to the function value.

We can make this more specific below.

Theorem 6.2.1 **Volume by Disks for Rotation about the x-axis**

To find the volume of a solid formed by rotating a function $y = f(x)$ on $[a, b]$ about the x-axis first note that the area of each cross-section is $A(x) = \pi(f(x))^2$, then the volume is

$$V = \int_a^b A(x)\,\mathrm{d}x = \int_a^b \pi(f(x))^2\,\mathrm{d}x.$$

Exercise 6.2.1

Write a similar statement to the one above for rotating a function $y = f(x)$ for y values in $[a, b]$ about the y-axis. *Hint: careful with the variables you use in the expression. The variable of integration should be the variable used in the integrand.*

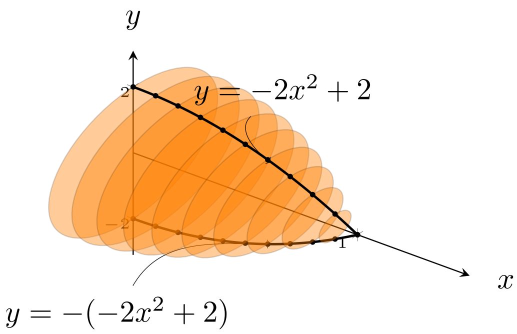
Example 6.2.1

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded above by $y = -2x^2 + 2$ around the x-axis.

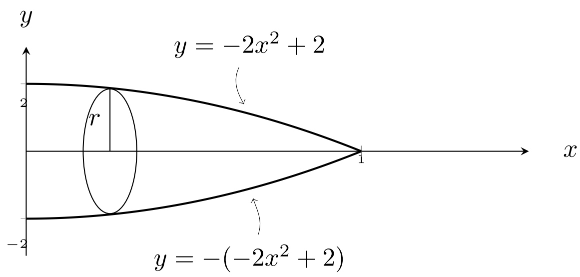
Set up an integral to find the volume when revolving this same region around the y-axis.

Solution

The solid looks like



However, this might be difficult to draw by hand. An alternative is to just draw the curve in 2-D, its reflection, and a single cross-section to help visualize the radius of the disks. This looks like



Then, we need to find where the curve hits the x-axis. Sometimes we can tell from the graph we draw, sometimes we have to solve for it

$$\begin{aligned} -2x^2 + 2 &= 0 \\ -2(x^2 - 1) &= 0 \\ -2(x+1)(x-1) &= 0 \\ x &= 1, -1 \end{aligned}$$

Since we are looking in the first quadrant, $x = 1$ is where this intersection happens (This is one of the bounds for the integral later). The area of the circular cross-section is $\pi r^2 = \pi(-2x^2 + 2)^2$. Thus, the integral we need is $V = \int_0^1 \pi(-2x^2 + 2)^2 \, dx$.

Exercise 6.2.2

Evaluate the integral above to find the volume of the solid.

Exercise 6.2.3

Set up and solve an integral to find the volume when revolving this same region around the y-axis.

Theorem 6.2.2 Volume by Disks for Rotation about the y-axis

To find the volume of a solid formed by rotating a function $y = f(x)$ (and solving for $x = f^{-1}(y)$) for x in $[a, b]$ and y between $f(a)$ and $f(b)$ about the y-axis first note that the area of each cross-section is $A(y) = \pi(f^{-1}(y))^2$, then the volume is

$$V = \int_{f(a)}^{f(b)} A(y) \, dy = \int_{f(a)}^{f(b)} \pi(f^{-1}(y))^2 \, dy.$$

In other words,

Note

Finding the Volume of a Solid of Revolution: Disk Method

1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radius of the disk.
5. Integrate. Remember units in your final answer when applicable.

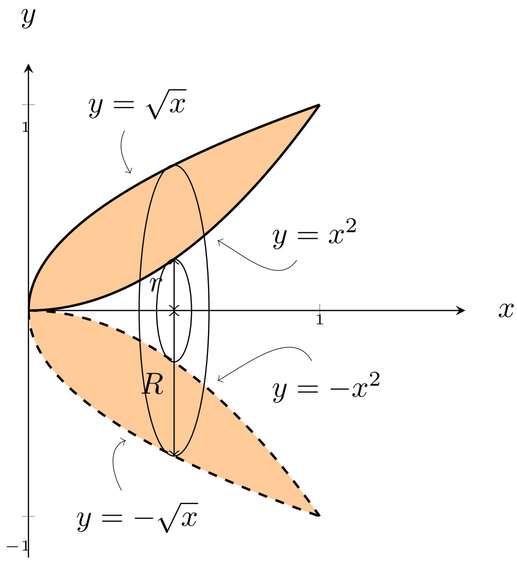
6.2.3. Solids of Revolution: The Washer Method

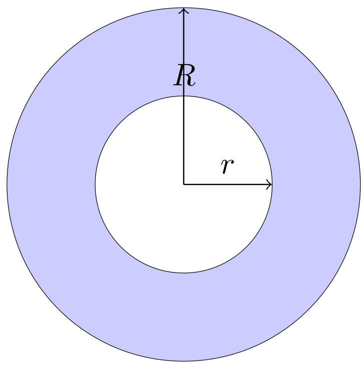
Note that we can only use the disk method if the cross-sections of the solid of revolution are circles. This means that there is no space gap between the region and the axis of revolution. What if there is a gap between the region and the axis of revolution?

Example 6.2.2

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded by $y = x^2$ and $y = \sqrt{x}$ around the x-axis.

Solution





We note the curves intersect at $(1,1)$. There are a few ways to think about this problem, but for now let's go back to thinking about the area of the cross-section and integrating that from 0 to 1 in x. For the cross-sections, the area is a circle with radius $R = \sqrt{x}$ minus the area of a smaller circle with area $r = x^2$. This region is sketched above in blue, and is known as a washer (as in the metal bit used with a nut and bolt that has the same shape). The area of the full region is the difference in areas between the 2 circles. That is, $A(x) = \pi R(x)^2 - \pi r(x)^2$. Then, integrating along the axis perpendicular to the cross-sectional area, we see the volume is

$$\begin{aligned}
V &= \int_0^1 A(x)\,\mathrm{d}x \\
&= \int_0^1 \pi R(x)^2 - \pi r(x)^2\,\mathrm{d}x \\
&= \pi \int_0^1 \left[\left(\sqrt{x}\right)^2 - \left(x^2\right)^2\right]\mathrm{d}x \\
&= \pi \int_0^1 x - x^4\,\mathrm{d}x
\end{aligned}$$

Exercise 6.2.4

Evaluate the integral above.

Theorem 6.2.3 **Volume by Washers for Rotation about the x-axis**

For a solid of rotation formed by rotating a region around the x-axis where you have the required bounds and radii in terms of x, the volume is

$$V = \int_a^b A(x)\,\mathrm{d}x = \pi \int_a^b \left[(R(x))^2 - (r(x))^2\right]\mathrm{d}x$$

Theorem 6.2.4 **Volume by Washers for Rotation about the y-axis**

For a solid of rotation formed by rotating a region around the y-axis where you have the required bounds and radii in terms of y, the volume is

$$V = \int_a^b A(y)\,\mathrm{d}y = \pi \int_a^b \left[(R(y))^2 - (r(y))^2\right]\mathrm{d}y$$

In other words,

ⓘ Note

Finding the Volume of a Solid of Revolution: Washer Method

1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radii of the washer: $R(x)$ and $r(x)$ (or $R(y)$ and $r(y)$ respectively).

- These are found by finding the distances of each curve from the axis of revolution. For more, see area between curves section.

5. Integrate. Remember units in your final answer when applicable.

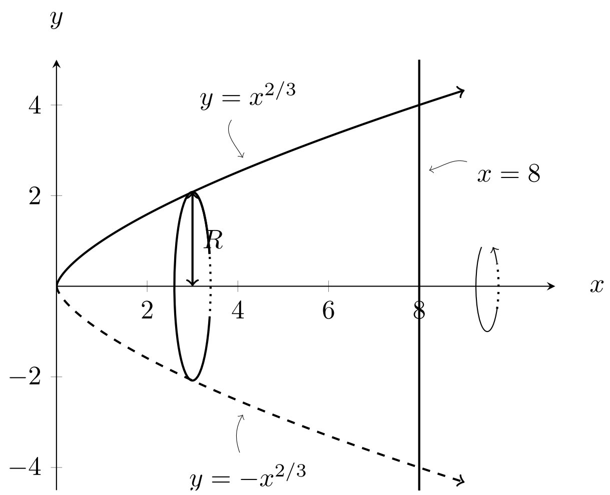
Example 6.2.3

Set up an integral to find the volume of the solid generated by revolving the region bounded by $y = x^{\frac{2}{3}}$, the x-axis, and $x = 8$

a) around the x-axis
b) around the y-axis
c) around the line $y = 6$.

Solution

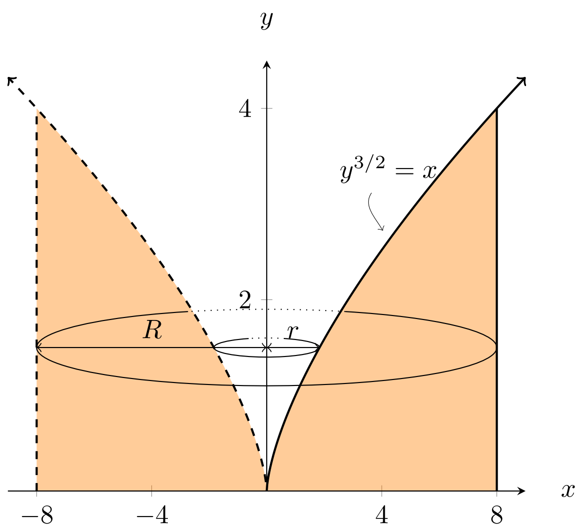
a) Sketching the region and an idea of the volume we make by rotating around the x-axis we see the cross-sections are disks.



Thus, the volume can be found with the disk method: by thinking of the cross-sectional area as a bunch of circles perpendicular to the x-axis, from $x = 0$ to $x = 8$.

$$
\begin{aligned}
V &= \pi \int_0^8 (R(x))^2 \,\mathrm{d}x \\
&= \pi \int_0^8 \left(x^{\frac{2}{3}}\right)^2 \mathrm{d}x \\
&= \pi \int_0^8 x^{\frac{4}{3}} \,\mathrm{d}x
\end{aligned}
$$

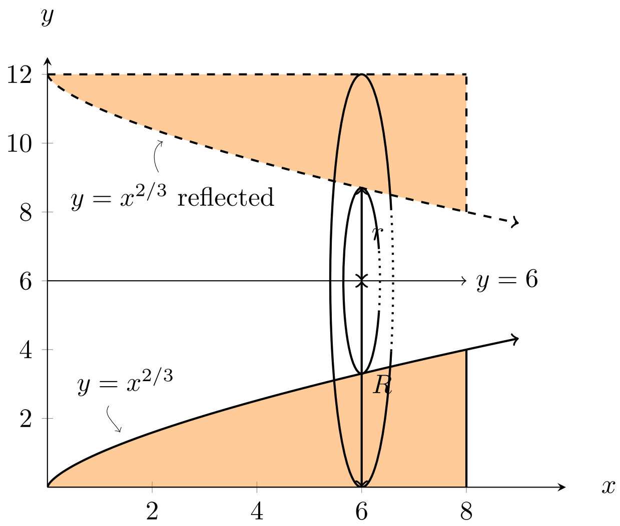
b) Sketching the region and an idea of the volume we make by rotating around the y-axis we see the cross-sections are washers now.



Notice that we need to express the radii in terms of y now, the large radius is always $R(y) = 8$ and the small radius is $r(y) = y^{\frac{3}{2}}$ which we got from solving the original equation for x. Lastly, we note the solid fills space for y in $[0, 4]$. Then the volume integral will be

$$\begin{aligned} V &= \int_0^4 \pi\left[(R(y))^2 - (r(y))^2\right] \mathrm{d}y \\ &= \int_0^4 \pi\left[(8)^2 - \left(y^{\frac{3}{2}}\right)^2\right] \mathrm{d}y \\ &= \pi \int_0^4 \left[64 - y^3\right] \mathrm{d}y \end{aligned}$$

c) Sketching the region and the horizontal line $y = 6$ we have



We first find the radii, or the distance from $y = 6$ to the edges of the region. Note R is always 6, and $r = 6 - x^{\frac{2}{3}}$. Then the area of each washer as a function of x is

$$\begin{aligned} A(x) &= \pi R^2 - \pi r^2 \\ &= \pi\left[6^2 - \left(6 - x^{\frac{2}{3}}\right)^2\right] \end{aligned}$$

The washers fill the space from $x = 0$ to $x = 8$, so the integral we use is

$$V = \pi \int_0^8 \left[6^2 - \left(6 - x^{\frac{2}{3}}\right)^2\right] \mathrm{d}x$$

6.2b Section Summary:

- We used the area of a cross-section to develop disk and washer method for finding volumes of the solids of rotation.
- We learned how to find the volumes of solids rotated about different horizontal and vertical lines.